

Daily Work Report

Project: Education Malaysia

Summary of Daily Activities

On September 11, 2026, development was focused on **Dynamic Sitemap generation & 404 URL Prevention, SEO Structured Data (JSON-LD) Google validation compliance, Breadcrumb Navigation & Content Retrieval fixes, JWT Authentication Middleware with refresh token cookie fallback, Page Content Repository integration, and Admin Portal (Vite + React 19) routing/APIs.**

Work Breakdown (Topic, Date, Description & Issues Fixed)

1. Dynamic Sitemap Generation & 404 URL Prevention Fixes

- **Topic:** Dynamic Sitemap Builder, Slug Duplication Handling & 404 Elimination
 - **Date:** 11 September 2026
 - **Files Modified:**
 - src/backend/services/sitemap-data.service.ts
 - src/lib/queries/specializations.ts
 - **Description & Issues Resolved:**
 - **Sitemap 404 URL Bug:** Sitemap generator mein aisi specialization pages aur level URLs publish ho rahi theen jinka backend content exist nahi karta tha, jiski wajah se search engine bots ko 404 Not Found mil raha tha. Service query mein EXISTS(SELECT 1 FROM specialization_contents WHERE ...) constraint enforce kiya taaki sirf valid pages sitemap mein aaein.
 - **Duplicate Slugs Resolution:** Database mein kuch specializations ke multiple entries hone ki wajah se non-deterministic row fetch ho rahi thi. Query ko stable banane ke liye ORDER BY cs.id ASC aur MIN(cs2.id) pin kiya taaki live website aur sitemap dono exact same database record fetch karein.
 - **Sitemap Performance Optimization:** URL generation pipeline ko streamlined aur optimized banaya taaki high traffic crawls par memory overflow na ho.
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2. SEO Structured Data (Schema.org JSON-LD) & University Data Query Fix

- **Topic:** Rich Snippets, Aggregate Rating Calculation & SEO Data Retrieval
 - **Date:** 11 September 2026
 - **Files Modified:**
 - src/lib/seo/structured-data.ts
 - src/lib/queries/universities.ts
 - **Description & Issues Resolved:**
 - **Google Schema Validation Error:** Google Search Console par validation error aa raha tha "Multiple reviews without aggregateRating object" jab individual reviews exist karte the par overall aggregate rating missing thi.
 - **Derived Aggregate Rating Engine:** clampRating utility aur auto-calculating aggregateRatingValue logic implement kiya jo actual review objects se dynamic average score nikal kar Schema.org JSON-LD schema me inject karta hai.
 - **University Query Data Retrieval:** fetchUniversityFull query mein review_count aur average_rating parsing fields add kiye taaki frontend aur SEO meta dono mein accurate student reviews data reflect ho sake bina blank values ke.
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3. Breadcrumbs Resolution, Dynamic Content Retrieval & Spacing

- **Topic:** Deeply Nested Breadcrumbs, Dynamic Page Content Retrieval & Layout Spacing
- **Date:** 11 September 2026
- **Files Modified:**
 - src/backend/repositories/content.repository.ts
 - src/backend/config/constants.ts
 - src/components/Breadcrumb.tsx
- **Description & Issues Resolved:**
 - **Breadcrumb Path & Formatting:** Multi-level dynamic routes (e.g. Home > Universities > [Type] > [Slug] > Courses > [CourseSlug]) ke liye dynamic label retrieval theek kiya taaki raw slug ke bajaye clean, readable titles dikhein aur separator icon spacing balanced rahe.

- **Content Repository (static_page_contents):** Dynamic static page content fetch karne ke liye structured ContentRepository create kiya jisse pages ka hero text, meta tags aur body content seamlessly retrieve ho sake bina page break ya white screen issue ke.
 - **Centralized Constants:** Backend API endpoints, base URLs aur global configuration values ko constants.ts mein centralize kiya.
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4. JWT Authentication Middleware & Cookie Fallback

- **Topic:** Auth Middleware, Protected API Routes & Token Refresh Fallback
 - **Date:** 11 September 2026
 - **Files Modified:**
 - src/backend/middleware/auth.ts
 - middleware.ts
 - **Description & Issues Resolved:**
 - **Auth Session Lost Issue:** API calls ke dauran token expire hone par ya Authorization header missing hone par user logged-out ho raha tha.
 - **Cookie Fallback Implementation:** JWT verification middleware develop kiya jo Authorization header ke alawa HTTP-only refresh cookies se token fallback handle karta hai taaki user session stable rahe.
 - **Protected Endpoints Security:** Admin routes aur student portal endpoints ke liye role-based verification enforce ki.
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5. Admin App Portal Integration & Next.js Rewrites

- **Topic:** Admin App Setup (Vite + React 19), Admin Auth APIs & Next.js Routing
- **Date:** 11 September 2026
- **Files Modified:**
 - next.config.ts
 - admin-app/vite.config.ts
 - admin-app/src/App.tsx
 - src/app/api/v1/admin/auth/login/route.ts

- `src/app/api/v1/admin/auth/me/route.ts`
- `admin-app/package.json`
- **Description & Issues Resolved:**
 - **Admin App Architecture:** Admin panel ke liye separate Vite + React 19 + Tailwind setup initialize kiya.
 - **Next.js Proxy/Rewrites:** `next.config.ts` mein rewrite rules define kiye taaki `/admin/*` URLs Next.js se seamlessly internal Vite admin server par route ho sakein.
 - **Admin Auth**
Endpoints: Dedicated `/api/v1/admin/auth/login` aur `/api/v1/admin/auth/me` endpoints implement kiye admin credentials verify karne aur session state maintain karne ke liye.